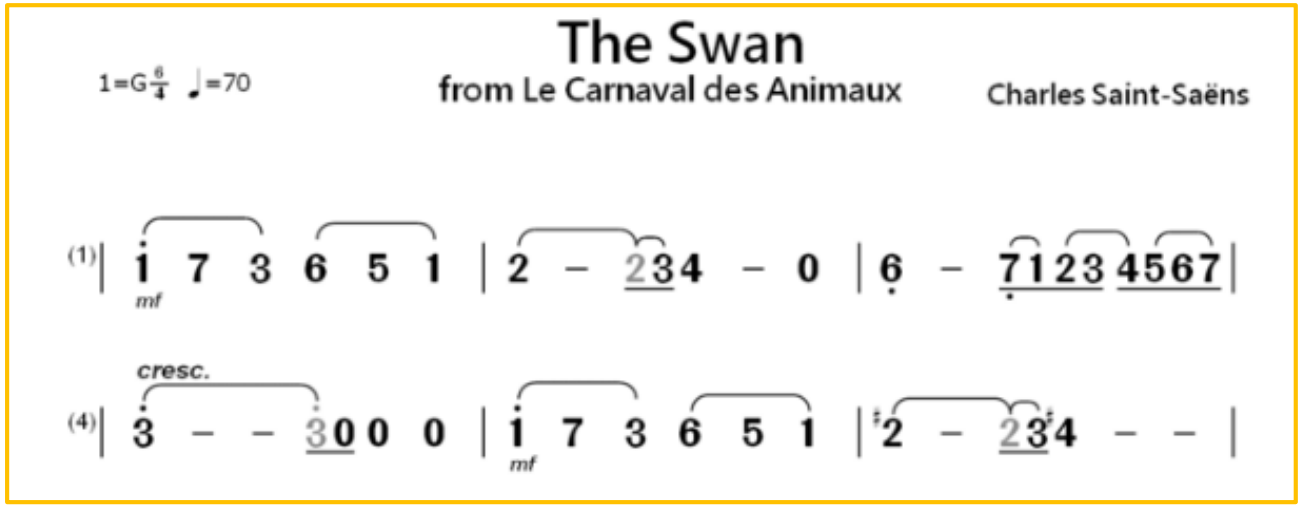
Project 2

Play the designated melody shown below with the buzzer installed on the 8051 A Plus board.

- ✓ Just play the notes as specified, the correctness of tempo is not required. However, the notes must be heard clearly.
- ✓ Extra bonus (5%) will be awarded for the correctness of tempo.

Setting:

- G4 = 1
- G5 = $\dot{1}$

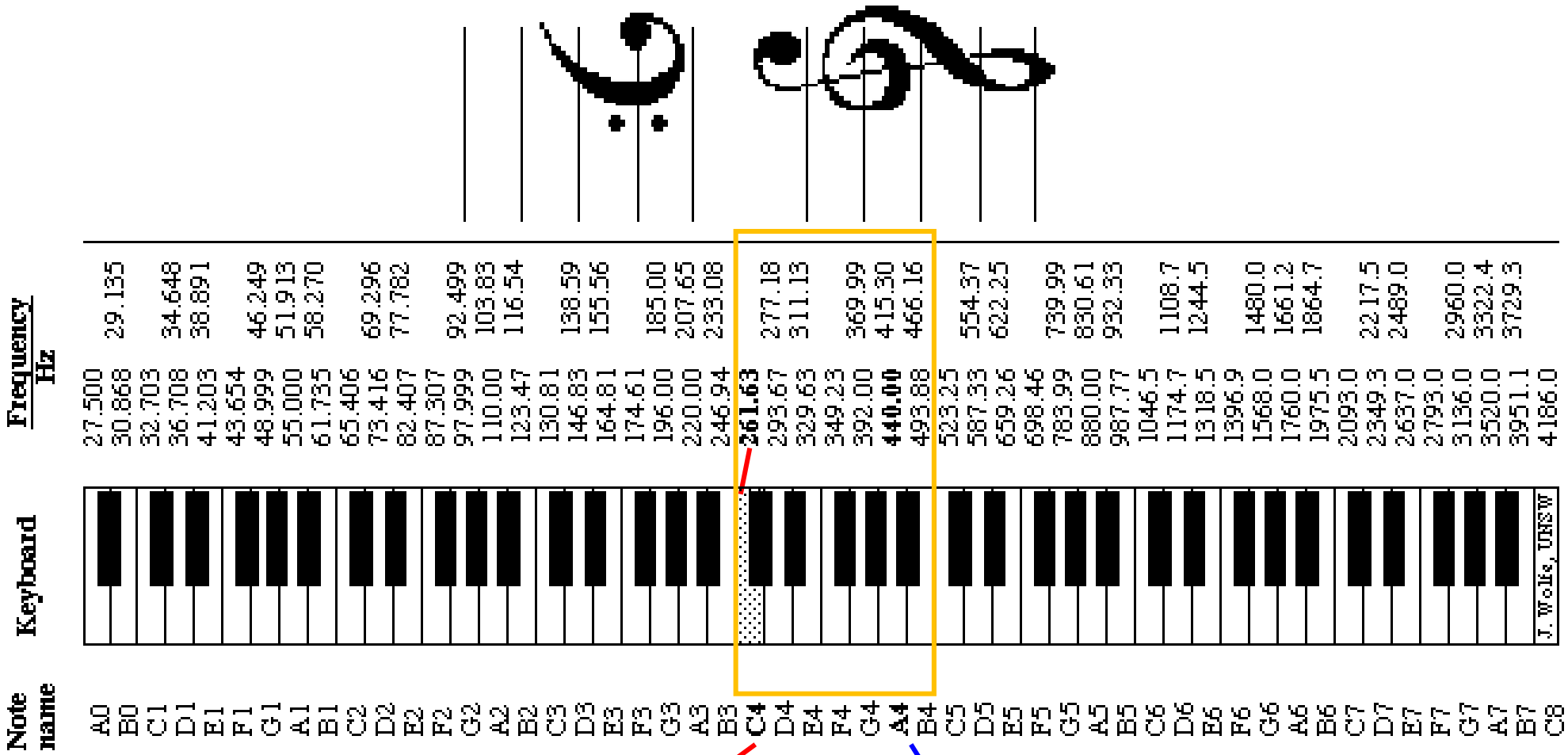


Note Frequency Chart

NOTE	OCTAVE 0	OCTAVE 1	OCTAVE 2	OCTAVE 3	OCTAVE 4	OCTAVE 5	OCTAVE 6	OCTAVE 7	OCTAVE 8
C	16.35	32.7	65.41	130.81	261.63	523.25	1046.5	2093	4186.01
C#/Db	17.32	34.65	69.3	138.59	277.18	554.37	1108.73	2217.46	4434.92
D	18.35	36.71	73.42	146.83	293.66	587.33	1174.66	2349.32	4698.63
D#/Eb	19.45	38.89	77.78	155.56	311.13	622.25	1244.51	2489.02	4978.03
E	20.6	41.2	82.41	164.81	329.63	659.25	1318.51	2637.02	5274.04
F	21.83	43.65	87.31	174.61	349.23	698.46	1396.91	2793.83	5587.65
F#/Gb	23.12	46.25	92.5	185	369.99	739.99	1479.98	2959.96	5919.91
G	24.5	49	98	196	392	783.99	1567.98	3135.96	6271.93
G#/Ab	25.96	51.91	103.83	207.65	415.3	830.61	1661.22	3322.44	6644.88
A	27.5	55	110	220	440	880	1760	3520	7040
A#/Bb	29.14	58.27	116.54	233.08	466.16	932.33	1864.66	3729.31	7458.62
B	30.87	61.74	123.47	246.94	493.88	987.77	1975.53	3951.07	7902.13

Unit: Hz

Note Frequency Chart



Middle C = C4 @ 261.63 Hz

Middle A = A4 @ 440 Hz

Materials to Submit for the Project

1. A text file of source code in .asm format. Try to make the code clear and elegant; the comment clear and succinct, if possible.
2. A video demonstration includes the entire designated section of melody.
3. A PPT report which contains:
   1) Flow chart.
   2) Discussion of your thought/idea regarding the project.
   3) Acknowledge the help received, if any.

DO NOT put segments of source code in your report, unless you have something special and would like to show off.

Grading Policy

1. 60%: successful implementation of the program.
2. 25%: power point presentation, including, but not limited to,
   a) Flow chart.
   b) Discussion.
3. 10%: style of program: succinct and structural program, comments are elegant and clear of information.
4. 5%: short video clearly show your project result.

Example Points for Discussion

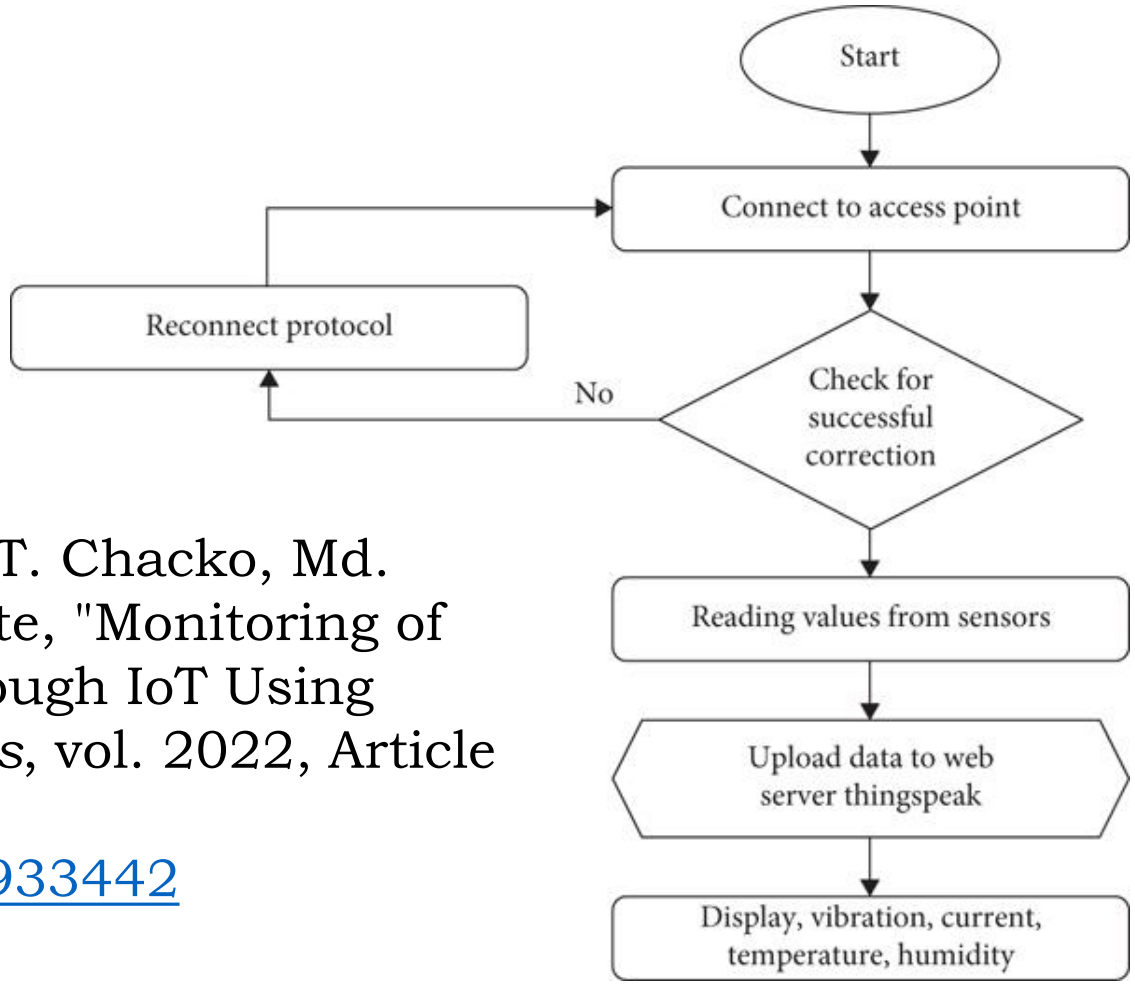
1. Is it possible to turn up/down the volume? If so, how? If not, why?
2. What is the lowest / highest note the buzzer onboard the 8051A plus board can play?
3. Is it possible for the buzzer to play more than 1 note, for example 3 notes C4, E4, and G4, concurrently? If so, make a demo. If not, why?
4. Difficulty encountered and idea to solve the problem.

The thought/idea to think about is not limited to those listed above.
It is a good habit to consider, based on the project content, whether it is possible to generalize the capability of the project outcome (and how), find different application aspects (may be with little modification), extend concepts involved in the project, or what can be done next, etc.

Points to Note

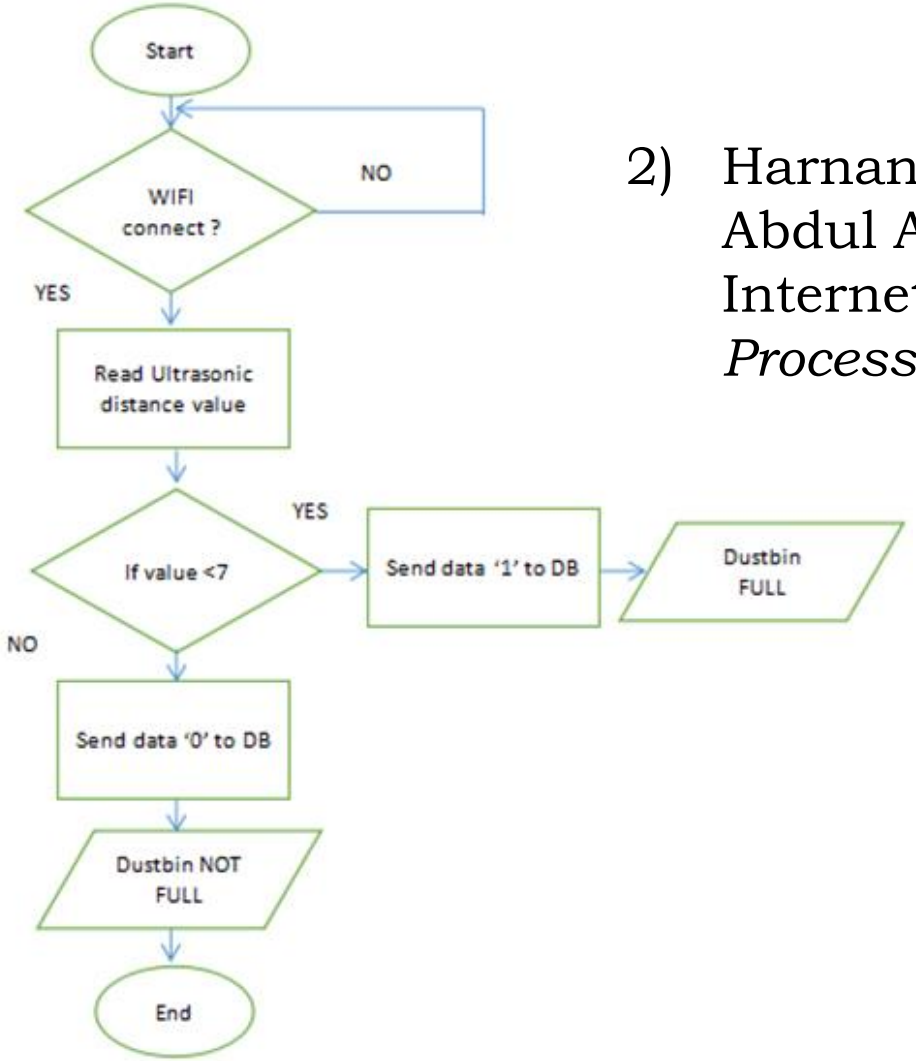
1. In case get stuck, you are welcomed and encouraged to discuss with your friends/classmates.
2. Bonus will be awarded to the groups that provide help. It is your responsibility to acknowledge the help received, if any, in your report. Give a brief description on the difficulty you have and the help you receive.
3. Don’t cheat. Don’t copy any result (code or report) of other group. Once caught, all groups involved will be failed for this project.
4. To avoid your work being stolen, don’t give your report to other group.

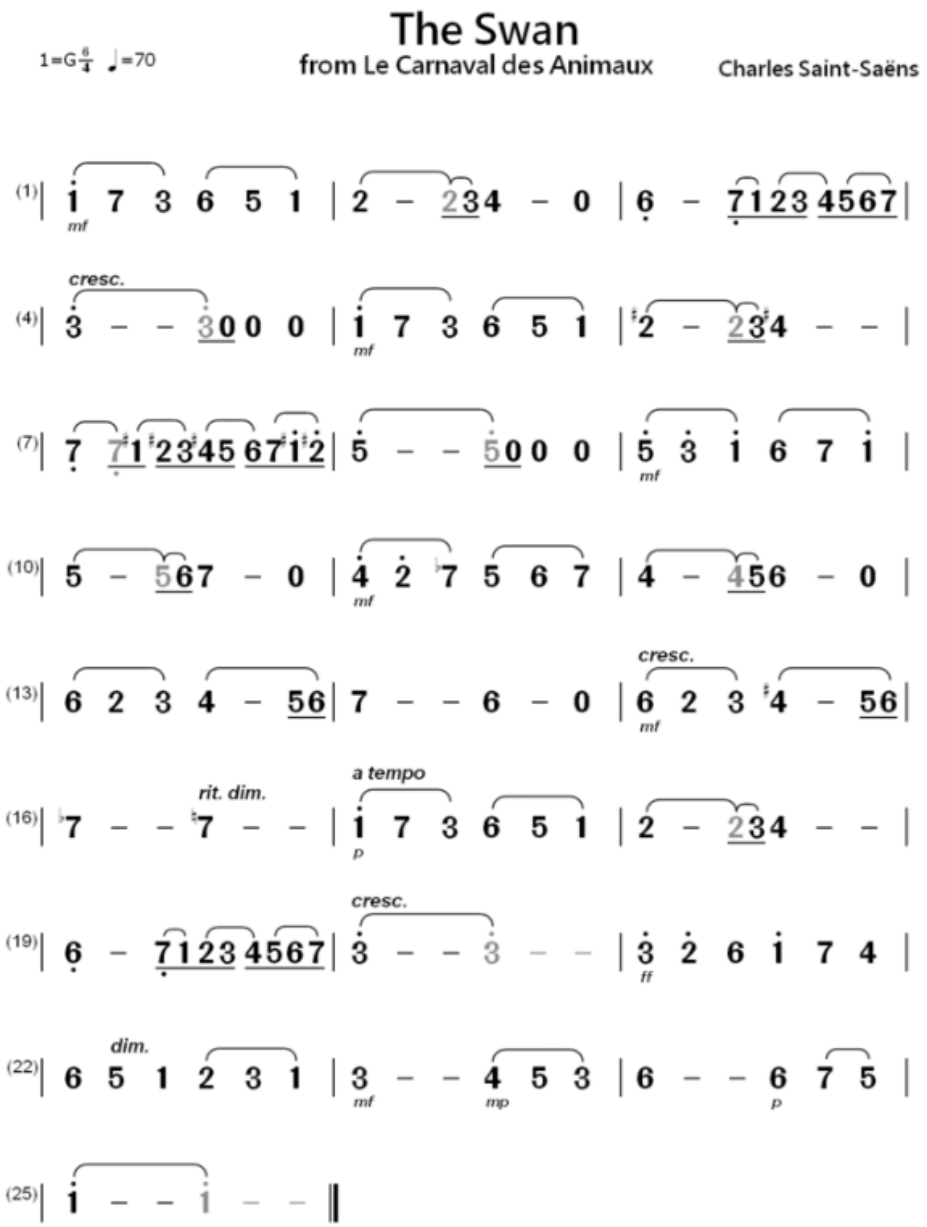
Reference – Example of Flowchart



1) Abhinab Shukla, S. P. Shukla, S. T. Chacko, Md. Khaja Mohiddin, Kinde Anlay Fante, "Monitoring of Single-Phase Induction Motor through IoT Using ESP32 Module", *Journal of Sensors*, vol. 2022, Article ID 8933442, 8 pages, 2022. https://doi.org/10.1155/2022/8933442

Reference – Example of Flowchart



2) Harnani Hassan, Fadzliana Saad and Nor Fazlin Abdul Aziz, "Waste Monitoring System based on Internet-of-Thing IoT", *IEEE Conference on Systems, Process and Control, ICSPC 2018*

In case you are interested in playing the whole song, the sheet music is here for your reference.